

Bangalore Ride Analytics: Demand, Cancellations & Trip Insights

Business Analysis Report

WHAT THIS PROJECT DOES

Uses SQL, Python, statistics and Power BI to turn 50,000 ride records into practical marketplace insights.

RIDES	COMPLETION	CANCELLATION	AVG. FARE
50,000	89.9%	10.1%	₹547.39
records analysed	overall rate	overall rate	completed rides

Source: Bangalore Rapido Ride Services Dataset

01 Executive Summary

This project looks at ride-level data from Bangalore to understand three practical marketplace questions: where demand comes from, where cancellations occur, and how trip economics change across different ride segments.

The analysis covers 50,000 rides. Of these, 44,964 were completed and 5,036 were cancelled, giving an overall completion rate of 89.9% and cancellation rate of 10.1%.

The clearest findings are straightforward: Bike is the highest-volume service; weekday demand is much higher than weekend demand; hourly demand is fairly stable; cancellation rates stay close to the 10% benchmark across most segments; and fare per kilometre falls sharply as trip distance increases.

At a glance

COMPLETED	CANCELLED	AVG. DISTANCE	AVG. DURATION
44,964	5,036	25.50 km	64.37 min
rides	rides	completed rides	completed rides

02 Project Objective & Business Questions

The goal was not to build a complex predictive system. Instead, the project follows a practical analyst approach: validate the numbers, find meaningful patterns, communicate them clearly, and turn the strongest patterns into questions that a marketplace team could investigate next.

Business questions

- How does ride demand vary across service types, hours, days and observed months?
- Where are cancellations concentrated, and are the differences large enough to justify investigation?
- How do fare, distance and duration behave across different trip segments?
- Which high-volume or higher-risk segments deserve deeper operational, supply or pricing analysis?
- What additional data would be needed to move from descriptive analysis to causal decisions?

03 Data & Analytical Approach

The cleaned dataset contains 50,000 ride records with fields for service type, date/time, ride status, source and destination, distance, duration, fare, payment method and several derived analysis fields.

Workflow

Stage	What was done	Main output
1. Prepare	Clean fields and create analysis-ready variables.	cleaned_rides.csv
2. Validate	Check counts, missing values, duplicates, types and basic ranges.	SQL data-quality checks
3. Analyse	Measure KPIs and compare service/time/trip segments.	SQL + Python
4. Test	Use statistical analysis to support group comparisons.	Python notebook
5. Communicate	Build a decision-oriented three-page dashboard.	Power BI

Data quality notes

- 44,964 records are completed and 5,036 are cancelled.
- Fare-related fields are missing on the cancelled rides, so fare and revenue analysis should use the available-fare population.
- The analysis includes derived fields such as hour, day of week, month, time period, route and distance bucket.
- The project uses completed rides consistently for the main trip-economics metrics.

04 Demand & Marketplace Performance

Service mix

Bike is the highest-volume service in the observed dataset, making it the most natural starting point for a deeper look at supply availability, service quality and customer experience.

Hourly demand

Ride volume stays within a relatively narrow range of about 2.0K–2.2K rides per hour. The highest observed hourly volume is around 9 AM. The dataset therefore shows variation by hour, but not an extreme single-hour concentration.

Weekday vs. weekend

A CLEAR DEMAND PATTERN

Weekdays account for 36,876 rides versus 13,124 on weekends. This is a meaningful difference in volume, although the dataset does not by itself establish why it occurs.

Observed monthly demand

The dataset contains June, July and August. July has the highest ride volume among the three observed months. Because only these three months are present, the result should not be treated as a full-year seasonal trend.

05 Cancellation & Operations

The cancellation analysis is deliberately cautious. Rates are close to the overall 10.1% benchmark across service, time period, distance and day, so the evidence does not point to one simple segment as the root cause.

Cancellation by service

Service	Cancellation rate	Read
Cab Economy	10.33%	Slightly above benchmark
Bike	10.32%	Slightly above benchmark
Bike Lite	10.16%	Near benchmark
Auto	9.84%	Near benchmark
Parcel	9.55%	Slightly below benchmark

Cancellation by time period

Time period	Cancellation rate
Evening Peak	10.28%
Daytime	10.12%
Late Night	10.09%
Morning Peak	10.05%
Night	9.78%

What this means

- Hourly cancellation rates fluctuate around 10%, with a few higher-risk hours.
- Distance-bucket cancellation rates are broadly similar; longer trips cannot be described as consistently higher-risk.
- Day-of-week rates are also stable: Thursday is highest at 10.36% and Sunday is lowest at 9.54%.
- The strongest next step is to connect these patterns to operational data such as supply availability, ETA and captain acceptance.

06 Trip Economics

AVG. FARE	AVG. DISTANCE	AVG. DURATION	FARE / KM
₹547.39	25.50 km	64.37 min	₹43.76
completed rides	completed rides	completed rides	completed rides

Fare by time period

Average fare is highly stable across time periods, ranging from approximately ₹545.31 to ₹548.20. This means the dataset does not show a large time-of-day difference in average recorded fare.

Fare per KM by distance

Distance bucket	Average fare / KM	Business read
Short (≤3 km)	₹300.11/km	Highest unit fare
Medium (3–7 km)	₹115.88/km	Sharp step-down
Long (7–15 km)	₹52.62/km	Further decline
Very Long (>15 km)	₹18.83/km	Lowest unit fare

STRONGEST ECONOMICS SIGNAL

Fare per KM decreases sharply as trip distance increases. This points to different economics across trip-length segments, but it is not evidence of profitability because cost data is not available.

Distance and duration

Average trip distance remains broadly around 25–26 km across time periods, while average duration stays around 64–65 minutes. The differences are small and should not be over-interpreted.

07 Statistical Analysis

The statistical notebook is used as a supporting layer to the dashboard. Its purpose is to examine whether observed group differences are worth further attention, rather than to turn descriptive patterns into causal claims.

- The analysis uses the cleaned ride-level dataset and derived business segments.
- Statistical testing is treated as supporting evidence for prioritization.
- A significant difference, by itself, would still not show that a business factor causes cancellation or demand changes.
- For causal questions, a controlled experiment is more appropriate.

08 Business Recommendations

01 Diagnose cancellation drivers

Use the 10.1% overall cancellation rate as the benchmark. Investigate segments that consistently sit above it, then connect those patterns to supply, ETA, captain acceptance and cancellation-reason data.

02 Start with high-volume services

Bike is the largest observed service by ride volume. Prioritize it when evaluating supply availability, matching quality and customer experience.

03 Investigate distance-based economics

The large decline in fare per KM across distance buckets makes trip length a strong candidate for deeper pricing and unit-economics analysis.

04 Plan around day type

Weekday demand is much higher than weekend demand. Compare supply and service quality by weekday/weekend before deciding what operational response is needed.

05 Improve the data before adding model complexity

Additional operational variables are likely to create more value than adding a complex predictive model to the current dataset.

09 Experiment Opportunity

A natural next step is an A/B test of one specific operational or product intervention designed to reduce cancellations. The historical dataset can generate the hypothesis, but it cannot prove the intervention will work.

Element	Suggested design
Control	Current experience / operations
Treatment	One clearly defined cancellation-reduction intervention
Primary metric	Cancellation rate
Secondary metrics	Completion rate; booking-to-completion conversion
Operational metrics	ETA/time-to-match and captain acceptance, if available
Guardrails	Customer experience, support contacts and economics
Decision rule	Pre-agree practical improvement, sample size/power and rollout criteria

10 Power BI Dashboard

Page	Focus	Key evidence
01 – Marketplace Overview	Demand & service mix	50K rides, completion/cancellation KPIs, service volume, hourly demand, weekday/weekend and revenue.
02 – Cancellation & Operations	Cancellation diagnostics	Cancellation benchmark by service, hour, time period, distance and day.
03 – Demand & Trip Economics	Completed-ride economics	Fare, distance, duration, fare/KM, monthly demand and distance-segment economics.

11 Limitations

- Cancellation reasons are not available.
- Captain acceptance, supply availability and ETA/time-to-match are not available.
- Customer and captain behavioural history is limited.
- Recorded fare is not the same as profit or contribution margin.
- The analysis is observational; associations do not establish causality.
- Only June–August are present in the observed monthly data.

12 Conclusion

This project demonstrates a practical Business Analyst workflow from data preparation to business communication. The analysis identifies a clear demand pattern, a measurable cancellation benchmark and a strong distance-based fare pattern.

The main opportunity is not to make the current analysis more complicated. It is to use these findings to ask better operational questions, collect the missing supply-side data, and test targeted interventions with clear success metrics.

NOTE

This project created to demonstrate SQL, Python, statistics, Power BI, KPI analysis and business reasoning. It is not presented as internal company analysis.

